```
        Cylinder cylinder = new Cylinder(cylinderRadius, cylinderHeight);


        System.out.println("\nSphere Details");
        System.out.println("----------------------");
        sphere.area();
        sphere.volume();


        System.out.println("\nCylinder Details");
        System.out.println("----------------------");
        cylinder.area();
        cylinder.volume();


        sc.close();
    }
}
```

Slip 7

1)Write a package game which will have 2 classes Indoor & Outdoor. Use a function display () to generate the list of players for the specific game. Use default & parameterized constructor.

Step 1: mkdir -p javaproject/game

cd javaproject

Step 2:vim package/Indoor.java

```
package game;
public class Indoor {
   String game;
   String[] players;
   public Indoor() {
      game = "Table Tennis";
```

```
    players = new String[]{"Rahul", "Amit", "Sneha", "Priya"};
  }
  public Indoor(String game, String[] players) {
    this.game = game;
    this.players = players;
  }
  public void display() {
    System.out.println("Indoor Game: " + game);
    System.out.println("Players:");
    for (String player : players) {
      System.out.println(player);
    }
  }
}
```

Step 3: vim package/Outdoor.java

```
package game;
public class Outdoor {
  String game;
  String[] players;
  public Outdoor() {
    game = "Cricket";
    players = new String[]{"Virat", "Rohit", "Rahul", "Hardik"};
  }

  public Outdoor(String game, String[] players) {
    this.game = game;
    this.players = players;
  }
```

```
    public void display() {
        System.out.println("Outdoor Game: " + game);
        System.out.println("Players:");
        for (String player : players) {
            System.out.println(player);
        }
    }
}
```

Step 4: vim mainprogram.java

```
import game.Indoor;
import game.Outdoor;

public class mainprogram {
    public static void main(String[] args) {
        Indoor i1 = new Indoor();
        Outdoor o1 = new Outdoor();

        i1.display();
        System.out.println();
        o1.display();
        System.out.println();
        String[] indoorPlayers = {"Ajay", "Vijay"};
        String[] outdoorPlayers = {"Sachin", "Virat", "MS Dhoni"};
        Indoor i2 = new Indoor("Chess", indoorPlayers);
        Outdoor o2 = new Outdoor("Football", outdoorPlayers);
        i2.display();
        System.out.println();
```

```
    o2.display();
  }
}
```

Step5: javac game/Indoor.java game/outdoor.java mainprogram.java

Step6: java mainprogram

2)Write a java program to define a class MyDate (day, month, year) with methods to accept and display MyDate object. Accept date as dd,mm,yyyy. Throw user defined exception "InvalidDateException" if the date is invalid. Examples of invalid dates: 12 15 2015, 31 6 1990, 29 2 2001

```
import java.util.Scanner;
class InvalidDateException extends Exception {
  public InvalidDateException(String message) {
    super(message);
  }
}
class MyDate {
  int day, month, year;
  void accept() throws InvalidDateException {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter date (dd mm yyyy): ");
    day = sc.nextInt();
    month = sc.nextInt();
    year = sc.nextInt();
    if (month < 1 || month > 12)
      throw new InvalidDateException("Invalid month!");
    int maxDays;
    switch (month) {
```

```
            case 2:
               if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
                  maxDays = 29;
               else
                  maxDays = 28;
               break;
            case 4:
            case 6:
            case 9:
            case 11:
               maxDays = 30;
               break;
            default:
               maxDays = 31;
         }
         if (day < 1 || day > maxDays)
            throw new InvalidDateException("Invalid day!");
      }
      void display() {
         System.out.println("Date: " + day + "/" + month + "/" + year);
      }
   }
   public class DateDemo {
      public static void main(String[] args) {
         MyDate d = new MyDate();
         try {
            d.accept();
            d.display();
```

```
    } catch (InvalidDateException e) {
        System.out.println("Exception: " + e.getMessage());
    }
  }
}
```

OR

2) Write a java program to define a class student having rollno, name and percentage. Define Default and parameterized constructor. Overload the constructor. Accept the 5 student details and display it. (use this keyword).

```
import java.util.Scanner;
class Student{
int rollno;
String name;
float percentage;
Student(){
rollno=0;
name="Unknown";
percentage=0;
}
Student(int rollno,String name,float percentage){
this.rollno=rollno;
this.name=name;
this.percentage=percentage;
}
void display(){
System.out.println(rollno+"\t"+name+"\t"+percentage);
}
}
public class StudentDemo{
```